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Geochemical Signature of Au-Cu Mineralization in Soil Profiles of the Wadi Al Jaww Prospect and its Applications for Exploration, Saudi Arabia

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Abstract

A geochemical soil survey was carried out over the Wadi Al Jaww prospect located in the western part of the Arabian Shield between the Hijaz and Jeddah terranes. The area is approximately 150 km southwest of Maaden's Mahd Ahd Dahab mine and 150 km northeast of Jeddah city, and 200 km south of Madinah city. No historical work was reported at Wadi Al Jaww. The Maaden discovery was made from exploration geochemistry methodologies including silt, soil and rock samples. The gold and copper mineralization are hosted by the Shiwan complex of tonalite to granodiorite composition belonging to Kamil suite. The soil samples were collected from the horizons of a soil profile with a 50 m × 100 m grid spacing over the central part of the prospect and with a 100 m × 200 m spacing over northern and eastern extensions. The analytical results have been statistically examined using descriptive statistics, bivariate correlations, cluster and principal component analysis to determine elemental associations. Threshold values have been calculated through median±2MAD approaches to avoid the influence of outliers and to define subtle anomalies. PC-analysis was performed over CLR-transformed data and revealed associations of elements controlled by hypogene mineralization, the composition of host rocks, and surface processes. The first PC explains 47% of data variation and outlines the association of Au-Te. Gold is the main indicator of mineralization in soil profiles. Gold concentrations reach up to 2.39 ppm, indicating the position of near proximity to mineralization. While gold may be mobile in arid regions where the dissolution of elemental Au occurs by oxic, acidic, chloride-rich waters, the close spatial association of soil anomalies and hypogene gold mineralization indicates limited mobility of gold over the Wadi Al Jaww prospect. Tellurium is highly enriched in soils over mineralized zones but shows less consistent anomalies. The absorption of tellurium in soils could be associated with Fe (hydr) oxides and clay minerals. Copper is believed to be highly mobile in arid environments and usually incorporated into Fe-, Mn oxides and hydroxides. The composition of soil profiles and the arid environment at Wadi Al Jaww area explains the distribution of metals in soils.

The soil geochemistry survey was the main contributor for discovery of Wadi Al Jaww Au-Cu mineralization and outlined association of elements could be used as a guide for future discoveries of similar types of mineralization associated with intrusive complexes along suture zones of Arabian Shield.

Introduction

Soil sampling serves as a good tool in identifying potential areas of mineralization prior to other exploration activities. Geochemical analyses of surficial samples (stream sediments, soil, and rock samples) are performed to mitigate initial high risks, t he reby informing the planning of subsequent geophysical surveys and drilling campaigns, which are often cost-intensive components of the overall project. Preliminary results from surface studies serve as a base for further research and exploration activities within a specific exploration license area and have played a significant role in the discovery of currently operating deposits. Soil geochemistry helps to narrow down the exploration areas by identifying regions that should be prioritized within a relatively large area. The application of soil geochemistry for tracking potential mineralization, defining geochemical halos, and guiding deeper exploration has been well-reported. This method is particularly important in the discovery of near-surface or outcropping mineralization, such as that observed in the Wadi Al Jaww deposit. In addition, such studies and their statistical analyses of intrusion-hosted Au-deposit indicate that soil sampling and the subsequent statistical evaluations have proven to be effective tools in exploring gold deposits in the region.

Soil sampling conducted over granodiorite to tonalite complex of Al Kamil suite led to discovery of Au (+Cu, Mo) mineralized zones, special extensions of which were confirmed by further exploration (grab sampling and drilling) (https://www.maaden.com). The geochemical footprint of known Wadi Al Jaww target N-E oriented geochemical anomaly that extends to more than 3.5 km. Gold is the main indicator of mineralization in soil profile, other pathfinders are Ag, Cu, Mo, Bi, Te and partly Sn. In this study, we evaluate the geochemical data on collected soil samples. The assessment is based on the application of basic statistical tools such as clustering of selected elements and examining the relationships between the elements quantitatively with principal component analysis (PCA). Following the spatial analysis using threshold values, the results were utilized to generate anomaly maps using geographical information system (GIS). The results of the surficial geochemistry survey on the Wadi Al Jaww gold deposit are believed to be useful for exploration of similar deposits.

Regional and local geology

The Wadi Al Jaww deposit located at western part of Arabian-Nubian Shield (ANS) and hosted by Shiwan complex of tonalite to granodiorite composition belonging to Kamil suite. This suite consists of mafic, intermediate and felsic plutonic rocks of calc-alkalic and locally trondhjemitic affinities widely exposed in the southwestern part of the Jeddah terrane between the Bi'r Umq suture on the north and the Makkah area on the south. The suite intrudes already deformed layered rocks of the Samran and Zibarah groups and plutonic rocks of the Makkah batholith. Shrimp dating yields crystallization ages of 824±7 Ma and 802±5 Ma for the suite (Johnson, 2003).

The Shiwan complex consists of tonalite, diorite and monzogranite and extends over 85 km along the north-western border of Jeddah terrane (figure 1).

Ongoing drilling at Wadi Al Jaww project reveals that the main gold bearing structures are weakly laminated quartz veinlets and thin chlorite-chalcopyrite veinlets mostly developed next to the contact between granodiorite and porphyritic granites. Crosscutting relationships indicate that quartz veinlets are later than thin chlorite-sulphide veinlets. Additionally, a significant portion of the chlorite- sulphide veinlets could be reopened by later events. Chalcopyrite is the main copper bearing mineral in chlorite-sulphide veinlets, however geochemical data indicate present of tellurides and Bi-bearing minerals (anomalous values of Te, Bi). The massive or weakly laminated quartz veinlets contain chalcopyrite, molybdenite and visible gold, however, contain less volume percent of sulphides in comparison with chlorite veinlets. Based on geochemical data, geological mapping, logging data, the Wadi Al Jaww is considered as a multiphase deposit with the first phase of mineralization caused by magmatic-hydrothermal event and later orogenic event which overprints early one and could remobilize some elements.

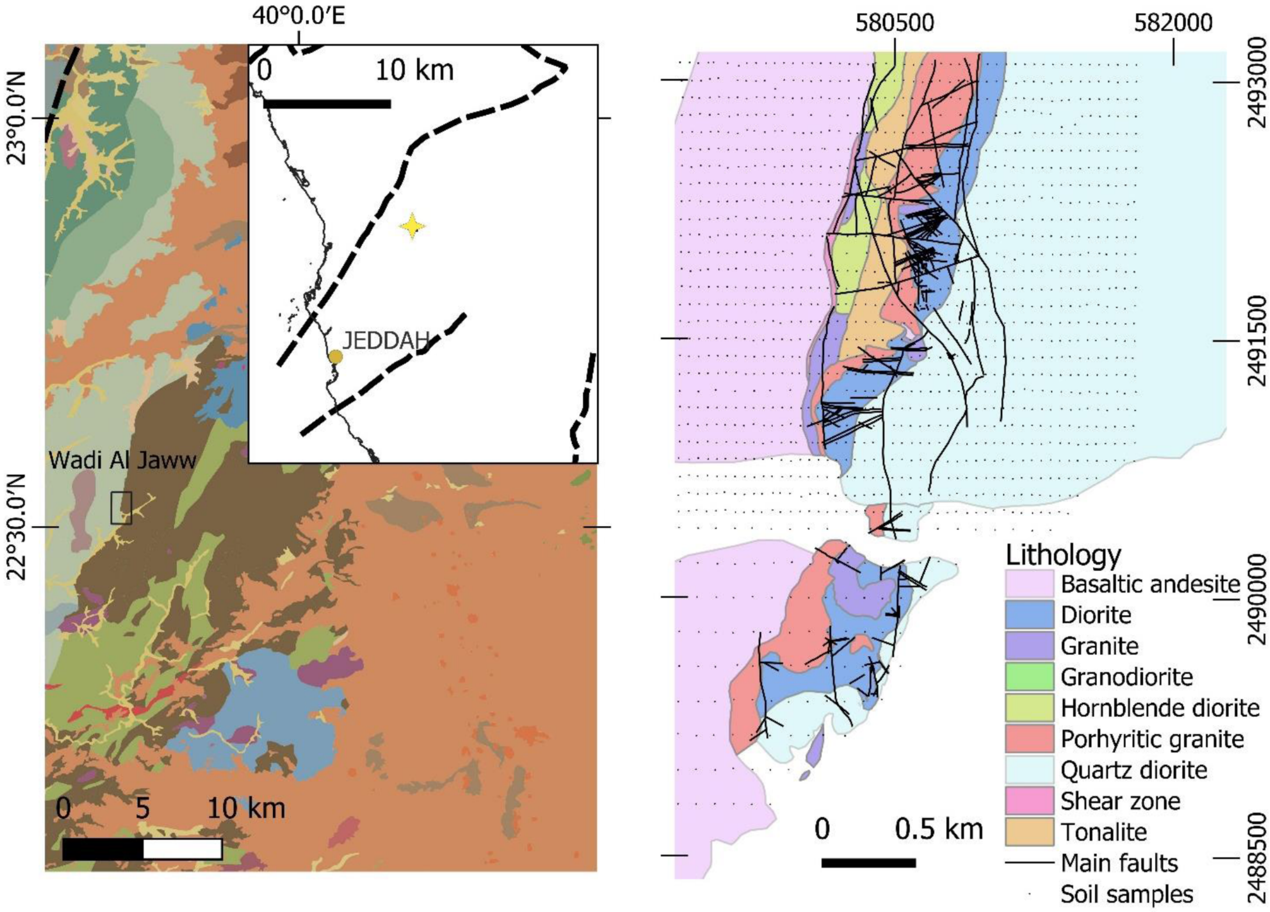

Figure 1—Location of the Wadi Al Jaww deposit and geological settings on regional and deposit scale.

Methodology

Soil sampling

The Wadi Al Jaww located in mountainous terrain with a poor development of soil profile. Soil samples were collected from "A" or "C" - horizons of soil profile and mainly represent residual material of weathered and disintegrated primary rocks. A total of 1441 samples were collected in a rectilinear grid at intervals of 50 × 100 m in the E-W direction of profiles. Collected material was sieved in the field to fraction −1 mm and then submitted to the laboratory. The coordinates in WSG84 UTM Zone 37 N were recorded with an accuracy of ±5 m. No samples were collected from areas with possible environmental pollution.

Analytical methods

Samples (1441 soil samples) were analyzed for 56 elements contents by ALS Arabia (Jeddah). Method code ME-MS41L was used for multi-element sample analysis and combines a two-acid aqua regia digestion of homogenized soil sample followed by analysis by ICP-MS or ICP-AES. A sample aliquot weighing c. 0.5 g is mixed with 75% aqua regia, 3:1 HNO_3:HCl, digested in a graphite heating block and made up to final volume of 12.5 ml with dH_2O. This method is particularly well suited for samples with high calcium content. Duplicate, blank samples and reference materials (Au dominant) were also used for quality assurance and quality control (QA/QC) procedures through the entire package of the collected samples. When the confidence interval used for evaluating the standard samples is defined as g ± 3 c, 90 % of the sample analyses satisfy the acceptance criteria. Duplicate samples were assessed against their corresponding original samples, and both the coefficient of determination (R^2) and the half absolute relative difference

(HARD) were evaluated. Reference materials and blank samples were deemed acceptable, as over 95 % of the assay results were either five times less than the lower detection limit or within the g ± 3c range.

Statistical analyses and geochemical mapping

Geochemical data were evaluated using ioGas (IMDEX) software for univariate and multivariate analyses. Descriptive statistics, normality tests and histograms were utilized to understand the probability features of data sets. Spearman's rho correlation, principal component analysis (PCA), and hierarchical clusters analysis (HCA) were used to evaluate the relationships among data quantitatively.

Spearman's rho (also called Pearson's rank correlation coefficient) is a statistical measure used to assess the strength and direction of the relationship between the two variables (from +1 to −1). Correlation values between 0 and + 1 indicate a positive linear relationship while the rest of the coefficients display negative correlations. Values between |0.00-0.19| show no or negligible relationship, |0.20- 0.29| weak relationship, |0.30-0.39| moderate relationship, |0.40-0.79| strong relationship and the values greater than |0.80| display very strong relationship (Prion and Haerling, 2014). Hierarchical cluster analysis was visualized with a dendrogram to group the analyzed elements, and the group linkage was used as a cluster method.

It is quite rare situation that raw data show a normal distribution, especially in geochemistry surveys conducted in mineral exploration stage (e.g., Yilmaz et al., 2015; Vural, 2019). Therefore, logarithmic transformation was applied to approach a normal distribution for soil samples. The transformed data were then used to determine threshold and anomaly values for the selected elements. Different procedures to identify data outliers in geochemical data were widely used and described in literature (Sinclair, 1974; Reimann et al., 2005). In exploration geochemistry values within the range [mean±2 sdev] were often defined as the 'geochemical background', but that empirical estimate is strongly influenced by extreme values. The other option that can mitigate the influence of outliers is using median absolute deviation instead of arithmetic mean [median±2 MAD]. Another solution would be to use the boxplot for the identifications of extreme values (Tukey, 1977). As demonstrated by Reinmann, 2005 the boxplot function is most informative if the true number of outliers is below 10%. If the proportion of outliers is above 15%, only the [median±2 MAD] procedure will perform adequately. Different methods of calculation threshold values were applied for analytical results of soil samples.

The QGIS software was utilized to generate maps based on the threshold values obtained from different methods. The kriging method was employed for interpolation in the geochemical mapping for the soils, as it is widely preferred for analyzing large datasets collected at equal intervals (e.g., Zuo, 2011; Shuguang et al., 2015; Vural, 2019).

The PCA method is a machine-learning algorithm that fundamentally reduces the dimensionality of large data sets by maximizing the variance of the given data (i.e., to two dimensions) (Joreskog et al., 1976; Howarth, 2013). When the examined data set is exceptionally large, which is the case in this study, PCA method is considered as an effective algorithm for further analysis (Grunsky and Smee, 1999; Cheng et al., 2006; Dempster et al., 2013). Performing the PCA method fundamentally requires that all variables have the same units of measure and data are certainly not needed to be normally distributed, but linear to use PCA. The components (PC) were extracted using the correlation matrix and rotated to an orthogonal simple structure using the variance maximizing (Varimax) criterion, which maximizes the variance of squared loadings of a variable across the components (e.g., Chandrajith et al., 2001; Tazikeh et al., 2018; Qavdar and Gunay, 2023). It attempts to make the factor loadings as large as possible for a small number of components, to make the relationship between the variables and the factors as interpretable as possible. This is the most common rotation for PCA when seeking to simplify component interpretation. Additional data transformations could be used to enhance recognition of anomalous multi-element associations. Modern approaches like CLR algorithm for opening lithogeochemical data and C-A fractal method for separating anomalies. Various methods have been introduced for transforming the composition data as log-ratio transformations. The most important of them are the Additive Log-Ratio transformation (ALR), the Centered Log-Ratio transformation (CLR), and the Isometric Log-Ratio transformation (ILR). The coefficients obtained from the logarithmic equations are the random variables that vary between®- and ® +. Therefore, it is easy to perform multivariate statistical analysis on them (Reimann et al. 2008). In the CLR method, the first logarithm of each variable will be specified, and then the logarithm is divided into the geometric mean of the variables. Here all variables must have the same unit. The one disadvantage of this method is the irreversibility of the covariance matrix of the variables (Khammar et al., 2021). The PCA method was performed for both untransformed and CLR-transformed data to see the difference between approaches and determine the paragenetic relationship of the elements.

Results

Descriptive statistical analyses, data distribution and transformation for soil samples

The Wadi Al Jaww prospect is considered as intrusion-hosted Au deposit with Ag, Cu, Bi, Te, Mo, Sb, As, Zn, Pb and Sn as main pathfinder elements. The highest concentrations in the analyzed soil samples of the raw data are as follows: Au 4.7 ppm, Ag 3.22 ppm, Cu 3110 ppm, Mo 44.8 ppm, Bi 19.85 ppm, Te 19.85 ppm, Sb 5.95 ppm, As 310 ppm, Zn 1605 ppm, and Pb 108.5 ppm. All elements have positive kurtosis and skewness values, and the variance is relatively high for Cu, Zn, As, and Pb (table 1). In addition to these parameters, outlier values of studied elements suggest a deviation from normal distribution. Therefore, logarithmic transformation was applied to minimize the effect of outlier values on the anomaly zones, which allowed us to make a much more reasonable data evaluation. Following the transformation of the data, the skewness and kurtosis were reduced to realistic levels, and a closer approximation to a normal distribution was achieved. Raw and transformed data of descriptive statistical parameters of the elements (Au, Ag, Bi, Cu, Mo, Bi, Te, Sb, As, Zn, Pb, Sn) from soil samples are given in Table 1. After logarithmic transformation all elements have positive skewness, while kurtosis is negative only for gold.

Table 1—Descriptive statistics of the raw and logarithmically transformed geochemical data of soil samples collected from Wadi Al Jaww area.

	Min.	Max.	Mean	Median	Std. Deviation	Variance	Kurtosis	Skewness
Au_ppm	0.003	4.70	0.07	0.01	0.24	0.06	129.07	9.46
Au_ppm_LOG	-2.602	0.67	-1.79	-1.89	0.71	0.50	-0.43	0.56
Ag_ppm	0.010	3.22	0.05	0.04	0.10	0.01	711.42	23.68
Ag_ppm_LOG	-2.000	0.51	-1.37	-1.41	0.23	0.05	5.89	1.59
Cu_ppm	11.350	3110.00	86.53	68.30	136.73	18694.20	243.81	13.76
Cu_ppm_LOG	1.055	3.49	1.84	1.83	0.23	0.05	8.32	1.59
Mo_ppm	0.140	44.80	1.38	1.23	1.33	1.77	790.69	24.60
Mo_ppm_LOG	-0.854	1.65	0.09	0.09	0.19	0.04	4.10	0.34
Bi_ppm	0.005	3.76	0.08	0.07	0.12	0.02	592.50	21.77
Bi_ppm_LOG	-2.284	0.58	-1.15	-1.15	0.20	0.04	8.74	1.16
Te_ppm	0.003	19.85	0.12	0.05	0.65	0.42	683.47	24.60
Te_ppm_LOG	-2.523	1.30	-1.28	-1.32	0.47	0.22	0.80	0.63
As_ppm	1.720	310.00	7.81	5.35	12.66	160.18	256.04	13.25
As_ppm_LOG	0.236	2.49	0.78	0.73	0.24	0.06	6.94	1.99
Sb_ppm	0.059	5.95	0.28	0.25	0.22	0.05	338.64	14.22
Sb_ppm_LOG	-1.229	0.77	-0.60	-0.61	0.19	0.04	2.96	0.78

	Min.	Max.	Mean	Median	Std. Deviation	Variance	Kurtosis	Skewness
Zn_ppm	24.000	1605.00	79.59	73.30	51.77	2680.24	537.63	19.48
Zn_ppm_LOG	1.380	3.21	1.87	1.87	0.14	0.02	10.12	1.42
Pb_ppm	1.555	108.50	5.70	5.36	3.81	14.49	407.94	16.98
Pb_ppm_LOG	0.192	2.04	0.73	0.73	0.14	0.02	9.93	1.16
Sn_ppm	0.240	6.78	0.84	0.82	0.29	0.08	137.24	7.87
Sn_ppm_LOG	-0.620	0.83	-0.09	-0.09	0.12	0.01	5.53	0.36

The Bivariate correlation and cluster analyses

The bivariate correlation was applied to measure the relationships among the analyzed elements. Pearson's rho coefficient was used for bivariate correlations. The correlation coefficients are presented in table 2. Gold exhibits a strong positive correlation with Ag, Cu and Mo as well as positive correlation with rest of pathfinder elements. The lowest positive correlation is between Au and Zn (0.029). Copper has a strong positive correlation with Bi, Au and moderate correlation with Te. The strongest correlation is between Ag and Mo (0.75), silver has a strong relationship with Te, As, Bi. Apart from that Sb, Zn and Sn do not correlate with any element. No significant strong negative correlation was observed, only subtle negative correlation displays between Sb and Zn (-0.06).

Table 2—Correlation coefficient for elements in the soil samples of Wadi Al Jaww area.

	Au	Cu	Mo	Bi	Te	As	Sb	Zn	Pb	Sn
Au	1									
Cu	0.52	1								
Mo	0.50	0.19	1							
Bi	0.35	0.63	0.18	1						
Te	0.33	0.48	0.46	0.60	1					
As	0.44	0.21	0.60	0.15	0.35	1				
Sb	0.18	0.10	0.14	0.09	0.05	0.31	1			
Zn	0.03	0.20	0.06	0.23	0.29	0.13	-0.06	1		
Pb	0.38	0.17	0.66	0.18	0.37	0.60	0.19	0.15	1	
Sn	0.25	0.20	0.22	0.13	0.07	0.22	0.12	0.02	0.31	1

	Au	Ag	Cu	Mo	Bi	Te	As	Sb	Zn	Pb	Sn
Au	1										
Ag	0.62	1									
Cu	0.52	0.49	1								
Mo	0.50	0.75	0.19	1							
Bi	0.35	0.51	0.63	0.18	1						
Te	0.33	0.72	0.48	0.46	0.60	1					
As	0.44	0.61	0.21	0.60	0.15	0.35	1				
Sb	0.18	0.13	0.10	0.14	0.09	0.05	0.31	1			
Zn	0.03	0.21	0.20	0.06	0.23	0.29	0.13	-0.06	1		
Pb	0.38	0.67	0.17	0.66	0.18	0.37	0.60	0.19	0.15	1	
Sn	0.25	0.22	0.20	0.22	0.13	0.07	0.22	0.12	0.02	0.31	1

Determination of threshold and spatial distribution

The normal abundance of any element in unmineralized and uncultivated soils (or the earth's surface) is generally considered as the natural background level (Reimann and Garrett, 2005). This can vary depending on the soil type and environmental conditions. On the other hand, the threshold represents the upper limit of the background range. It serves as a boundary value, that is values beyond which are considered anomalous, and values below which are regarded as a part of the background. Considering the inter-element relationships and the major components of mineralization as well as basic statistical parameters (e.g. mean, maximum, and standard deviation). Different approaches were tested to calculate threshold values for Au and pathfinder elements. Among the tested approaches, the logarithmic transformation demonstrated superior performance in stabilizing variance and reducing skewness, thus providing the most reliable normalization for subsequent analyses. The thresholds were determined based on the median and MAD calculated from the log- transformed data. Subsequently, the thresholds were back transformed by applying the antilogarithm to express them in the original (raw) data scale, in accordance with the standard practice of the method. The threshold values of the selected elements are given in table 3. Such approach provides a good result for all pathfinder elements; however, it results in high threshold values for gold (0.22 ppm).

Such a high threshold may lead to missing geochemical anomalies. The better results were achieved by calculating [median±2Robust standard deviation] calculated on raw data. Such approach provides a comparable result for most pathfinder elements with logarithmic transformed data and gives a more reliable threshold value for gold (0.088 ppm). Such a difference could be caused by the presence of (n=420) values below low detection limit (0.005 ppm). Analytical results with gold concentration below that value were replaced by half of that value (0.0025 ppm). Calculated threshold values were used to generate gridded maps of elements distribution over Wadi Al Jaww target.

Table 3—Threshold values in soil samples of Wadi Al Jaww calculated by calculated by different approaches.

	Au	Ag	Cu	Mo	Bi	Te	As	Sb	Zn	Pb	Sn
Mean	0.075	0.054	86.525	1.383	0.084	0.124	7.812	0.282	79.586	5.697	0.839
Median	0.013	0.039	68.300	1.230	0.071	0.048	5.350	0.246	73.300	5.360	0.820
sdev	0.240	0.100	136.727	1.330	0.124	0.649	12.656	0.216	51.771	3.806	0.287
Rsdev	0.037	0.019	22.387	0.497	0.023	0.066	2.520	0.099	17.902	1.342	0.178
MAD	0.067	0.019	24.025	0.367	0.020	0.090	2.962	0.076	13.786	0.923	0.121
Median +2MAD	0.146	0.077	116.351	1.963	0.111	0.229	11.273	0.398	100.871	7.205	1.061
Median+2sdev	0.494	0.239	341.753	3.890	0.318	1.346	30.662	0.678	176.842	12.972	1.394
Median+Rsdev	**0.088**	**0.076**	**113.075**	**2.223**	**0.117**	**0.180**	**10.391**	**0.444**	**109.105**	**8.044**	**1.176**
LogMedian +2MAD	0.222	0.073	107.411	2.110	0.110	0.241	9.972	0.418	102.796	7.526	1.092

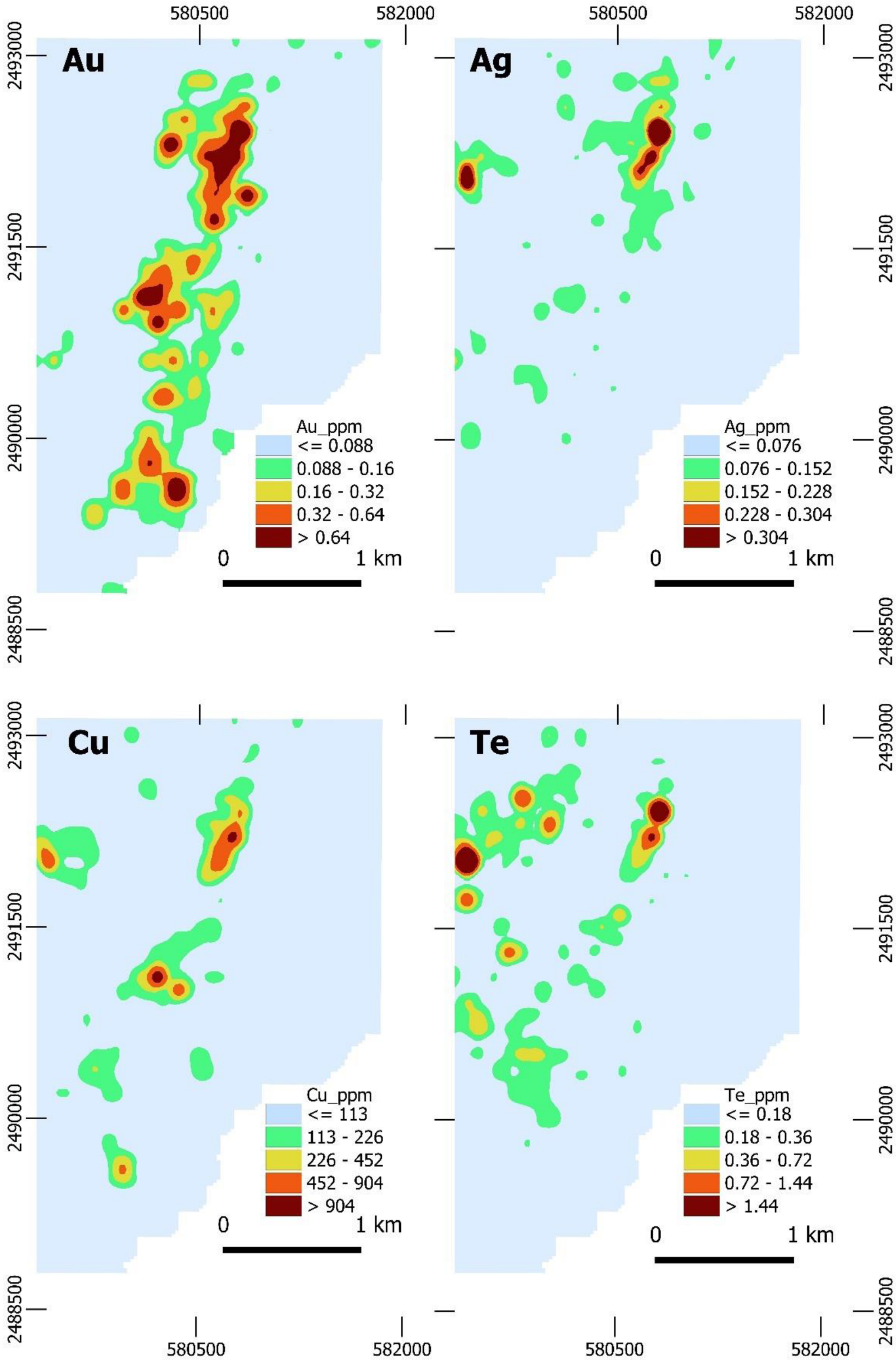

Figure 2—Distribution of elements in soil samples over Wadi Al Jaww deposit.

PCA method and data transformation

The PCA method was applied raw and CLR-transformed data of selected elements to reveal relationships between elements. Principal component analysis performed on raw data shows that PC1 has 40.56% of load, whereas PC2 has 14.58%. The last two principal components are responsible for 10 and 8.2% of variability of dataset, respectively (table 4). The PCA performed on CLR-transformed data shows different cumulative load of PC: PC1 - 31.92%, PC2 - 21.61%, PC3 - 11.31% and PC4 - 9.33%.

Table 4—Eigenvalues of PCA method for raw and CLR-transformed data

Raw data				CLR-transformed data			
Eigenvalues		Percent Cumulative %		Eigenvalues		Percent Cumulative %	
PC1	4.46	40.56	40.56	PC1	3.51	31.91	31.91
PC2	1.60	14.58	55.14	PC2	2.38	21.61	53.52
PC3	1.10	10.00	65.14	PC3	1.24	11.31	64.83
PC4	0.90	8.20	73.34	PC4	1.03	9.33	74.16

The eigenvectors for both raw and CLR-transormed data are showned in table 5. As it is seen from the table above the PC1 outlines association of Te-Mo-Ag-Pb-As-Au-Cu-Bi and has low values for Sb-Zn and Sn. At the same time PC2 shows association of As, Mo, Sb and has a negative load for Cu-Bi-Te-Zn, whereas doesn't show any load on Au and Ag. The PC3 outlines the association of Sb-Sn- Cu-Au and negative load for rest of the pathfinder elements. PC4 is mostly responsible for distribution of Sn (0.51) and has high negative load of Sb (-0.76). As seen from table 5, the PCA methods applied on CLR-transformed data show different association of elements. The PC1 has the strongest load for Au and Te, while the rest of pathfinder elements show negative or subtle values. The PC2 outlines association of Au-Sb-As-Mo and has a low or negative values for rest of pathfinder elements. The PC3 shows a high load on Cu and Ag and negative load on Mo-Bi-Te association. The PC4 on CLR-tranformed data shows Au-Mo-Ag+Mo association and negative load on rest of the pathfinder elements. The special distribution of PC calculated raw and CLR-transformed data showen on figure 3 and figure 4.

Table 5—PC loads for raw and CLR-transformed data

	Raw data					CLR-transformed data			
	PC1	PC2	PC3	PC4		PC1	PC2	PC3	PC4
Au_ppm	0.33	0.02	0.29	0.17	Au-CLR	0.43	0.28	0.05	0.26
Ag_ppm	0.44	0.00	-0.14	0.05	Ag-CLR	0.02	-0.29	0.49	0.29
Cu_ppm	0.29	-0.42	0.34	0.06	Cu-CLR	-0.06	-0.15	0.68	-0.10
Mo_ppm	0.36	0.30	-0.22	0.12	Mo-CLR	-0.29	0.23	-0.27	0.42
Bi_ppm	0.28	-0.49	0.18	-0.07	Bi-CLR	-0.22	-0.46	-0.24	0.02
Te_ppm	0.34	-0.29	-0.21	-0.09	Te-CLR	0.28	-0.42	-0.37	-0.25
As_ppm	0.33	0.33	-0.09	-0.19	As-CLR	-0.02	0.34	0.12	-0.71
Sb_ppm	0.12	0.25	0.49	-0.76	Sb-CLR	-0.22	0.42	-0.07	-0.11
Zn_ppm	0.13	-0.31	-0.46	-0.26	Zn-CLR	-0.38	-0.27	-0.03	-0.27
Pb_ppm	0.34	0.33	-0.21	0.04	Pb-CLR	-0.46	-0.03	0.03	0.02
Sn_ppm	0.17	0.17	0.39	0.51	Sn-CLR	-0.45	0.14	0.07	0.14

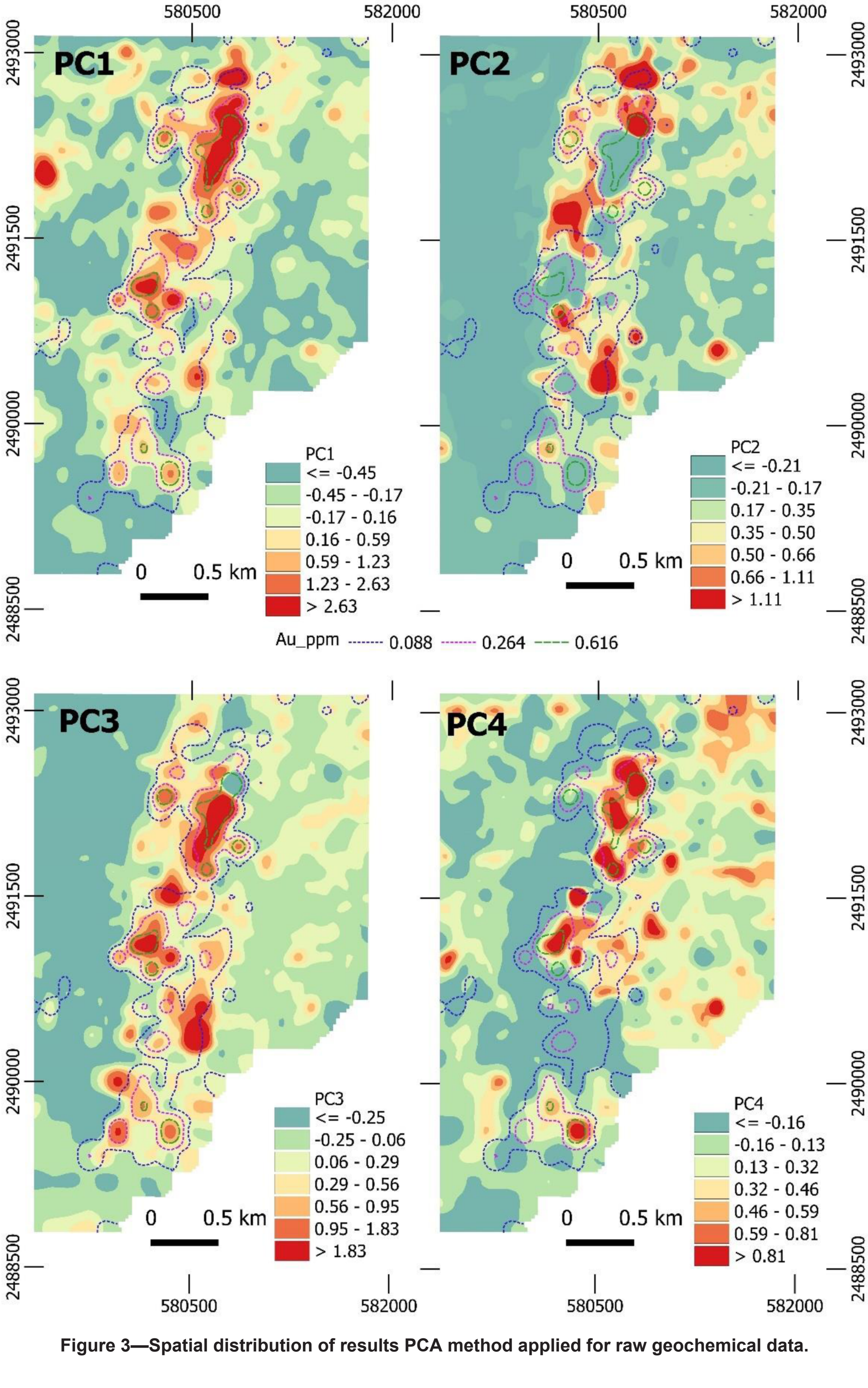

Figure 3—Spatial distribution of results PCA method applied for raw geochemical data.

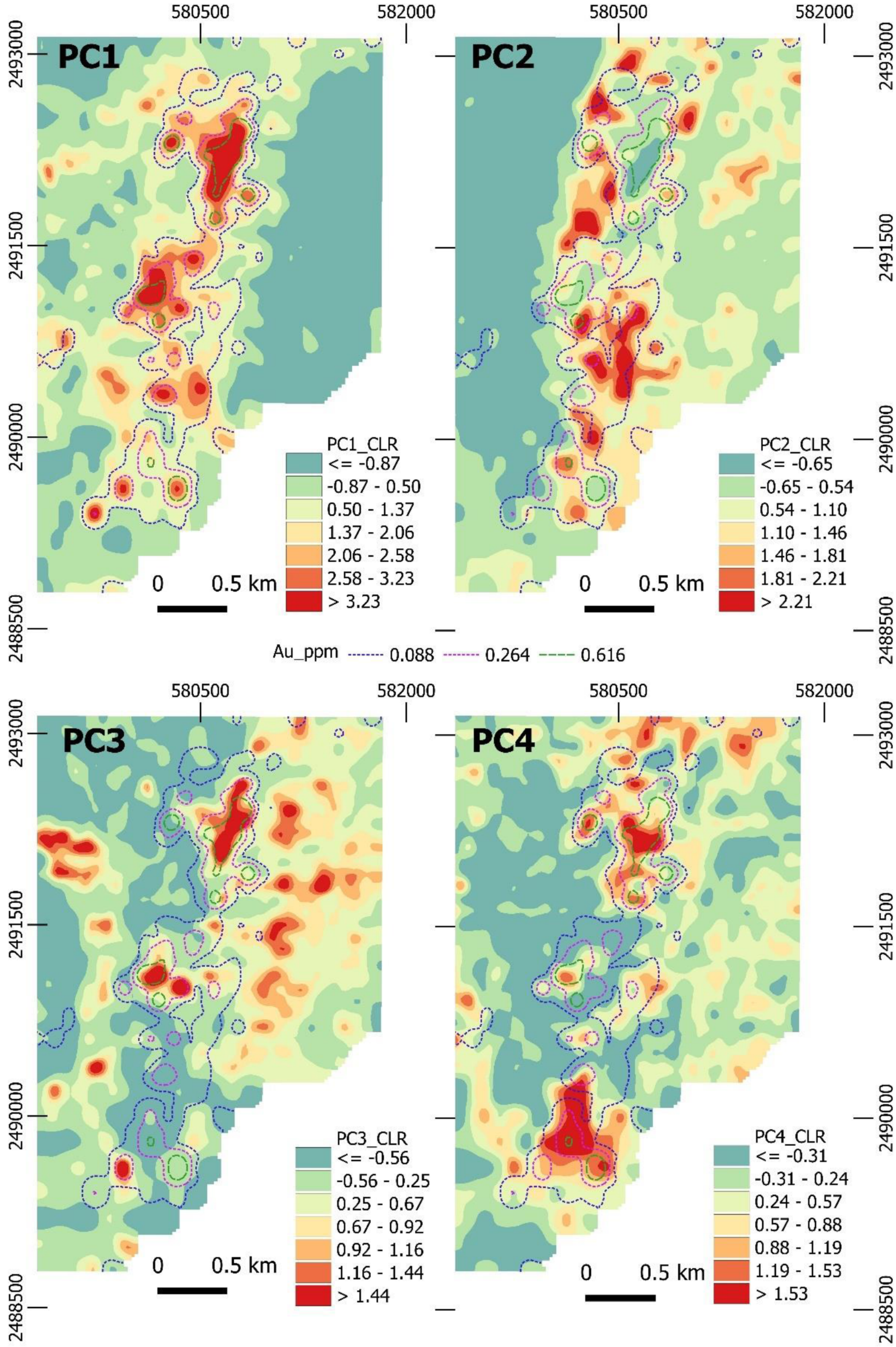

Figure 4—Spatial distribution of results PCA method applied for raw geochemical data.

Discussions and conclusions

The relationships between elements are frequently linked to local geological characteristics, the nature of the parent lithology, and soil formation processes (Anand and Butt, 2010; Tazikeh et al., 2018). The Wadi Al Jaww deposit is characterized by strong and widespread Au anomaly that outlined mineralization zone.

The other elements show less widespread anomalies and have a lower concentration. However, at the same time they show a good special relationship with gold. However, some special variations are present in the distribution of gold and other elements. Gold forms three separate clusters: northern, central and southern. The most anomalous is the southern cluster which is also characterized by presents of Ag and Te and subtle Cu anomaly. The central cluster is characterized by Au and subtle Cu anomaly, whereas southern cluster is characterized by only Au anomaly. Such differences could be explained by structural complexity of the deposit caused by late faults or decreasing the temperature of hydrothermal fluid reflecting transition from Au-Ag-Au-Cu-Te and Au only anomalies.

The correlation coefficients indicate the presence of a positive correlation between Au, Ag, Cu, Mo which can be explained by their immobile behavior against weathering of the pre-existing mineralization (Kuhn and Sigg, 1993). Such elements as Au, Ag, Mo believed to have limited mobility in arid environments however copper believed to be highly mobile in arid and acid environments and usually incorporates into Fe-, Mn oxides and hydroxides. Despite that, copper doesn't produce wide anomalies and seems not to be displaced far away from gold anomalies. That means that copper has also a limited mobility in such specific environments. The reasoning for that could be neutral or alkalic environments where copper is weakly mobile. The narrow anomalies of pathfinder elements should be explained by low content of sulfides, especially pyrite which is the min hosting mineral for many pathfinder elements. The absence of disseminated or veinlet-hosted sulfides (especially pyrite) could also explain 'in-suti' location of Ag, Te and Cu anomalies, because such settings are not able to produce enough acid to leach copper. Tellurium is highly enriched in soils over mineralized zones but shows less consistent anomalies. The absorption of tellurium in soils could be associated with Fe (hydr) oxides and clay minerals.

Performed comparison between differently calculated PCA on both raw and CLR-transformed data reveals various associations of elements. It is noticeable that only three principal components have a significant load on data variation in both methods calculated. The first PC calculated on raw data shows a broad anomaly and corresponding to Te-Mo-Ag-Pb-As-Au-Cu-Bi association. Whereas PC1 calculated on CLR-transformed data revials strong association is responsible for Au-Te association and has significantly narrow anomalies. In both cases, the PC2 is responsible for distal anomalies and mainly reviel association of pathdinder elements, Pb-Mo-As and Au-Mo-As-Sb, respectivley. The third PC calculated on raw data shows association of Au-Cu-Bi-Sb, whereas PC calculated on CLR-transformed data reveals Cu-Ag association. Distibution of PC3 from raw datsa coinside with distibution of au anomaly, which is caused by significant Au load, whereas PC3 calculated on CLR transformed data has wider distribution and reveals additional anomalous corridors to the west and east from the main anomalous zone. The approaches used show familiar spatial distribution of anomalies but are characterized by different associations of elements. It is noticable that CLR-transformation allow to limit the association of elements that could be used as a more precice way for drill targeting and revielialing associations of elements. In case of Wadi Al Jaww deposit there is a significant load of PC1 which is responsible for Au-Te association which could be used as a main proxy for futher exploration. Reveled associations of elements could be explained by different styles of mineralization, Au-Te could be explained by presence of gold-tellurides, while Au-Mo-As-Sb could be caused by presense of orogenic related styles of mineralization. Such two types of mineralization were observed in drillholes, where gold is mainly hosted by thin sheeted quartz-sulhide veinlets and late wider quartz and massive (mainly arsenopyrite) veinlets. Such diverse types of mineralization also have a different spacial distribution: the first PC is mainly developed within intrusive rocks, mostly diorites, while the second PC has higher values close to the contact with intrusive complex and host lithologies. It could be explained by structural control of orogenic mineralization which mainly developed next to the contact between intrusive complex and host lithologies.

The gold has significant and most solid anomalies over the Wadi Al Jaww deposit and should be used as a first proxy for exploration targeting. Other pathfinder elements have less developed anomalies and are usually characterized by relatively low values. Distibution of anomalies do not show a strong displacemed from gold anomalies, and even Cu show low mobility in such environments. Such narrow anomalies could be explained by absence of sulphide mineralization (mainly pyrite) which ois the main host mineral for pathfinder elements. Such assumption also explaines low mobility of pathfinder elements because the absence of pyrite doesn't facilitate to formation of acid environments. Perfomed PC analyses shows defferent association of elements, however CLR-transfeomerd approach reviel presence of gold in two association Au-Te and Au-Mo-As-Sb. Such associations could be explained by different mineralization types developed within the Wadi Al Jaww which is confirmed by drilling results.

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